

Cytotoxic response of breast cancer cell lines, MCF 7 and T 47 D to triphala and its modification by antioxidants.

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The cytotoxic effects of Triphala (TPL), an Indian Ayurvedic formulation with known anti-cancer properties, has been investigated on two human breast cancer cell lines differing in their p53 status. In vitro studies showed that MCF 7 with wild type p53 was more sensitive to TPL than T 47 D, which is p53 negative. TPL induced loss of cell viability was determined by MTT assay. After 72h incubation, the IC 50 values for MCF 7 was found to be approximately 8microg/ml and that for T 47 D was approximately 26microg/ml. Moreover, TPL inhibited the clonogenic growth of MCF 7 cells, which was significantly recovered by pifithrin-alpha, the p53 inhibitor. However, pifithrin-alpha, did not modify TPL induced cytotoxicity in T 47 D cells. Exogenous addition of antioxidants, glutathione (GSH) and N-Acetyl-Cysteine (NAC) inhibited the anti-proliferative ability of TPL in both MCF 7 and T47 D. Annexin-V and propidium iodide double staining of cells treated with TPL for 2h revealed that TPL induced significant apoptosis in both the cell lines in a dose dependant manner but magnitude of apoptosis was significantly higher in MCF 7 than in T 47-D cells. TPL was also found to induce dose and time dependent increase in intracellular reactive oxygen species in both the cell lines. Present results have demonstrated that MCF 7 and T 47 D cells exhibited differential sensitivity to TPL, which seems to be dependant on their p53 status. Inhibition of anti-proliferative ability of TPL by antioxidants suggests a role for TPL induced ROS in the induction of apoptosis. It is concluded that p53 status of cancer cells formed an important factor in predicting the response of cancer cells to prooxidant drugs.